

Suggested Mark Scheme

- 1 Plant cells have a cellulose cell wall outside the cell surface membrane. The cellulose microfibrils are embedded in a matrix of hemicellulose, pectin and proteins.

The cell surface membrane of a plant cell contains the enzyme, cellulose synthase, which synthesises microfibrils. It catalyses bond formation between monosaccharides in the cytosol. The monosaccharides pass through the enzymes and leave the cell as they are joined to form microfibrils. This is shown in Fig. 1.

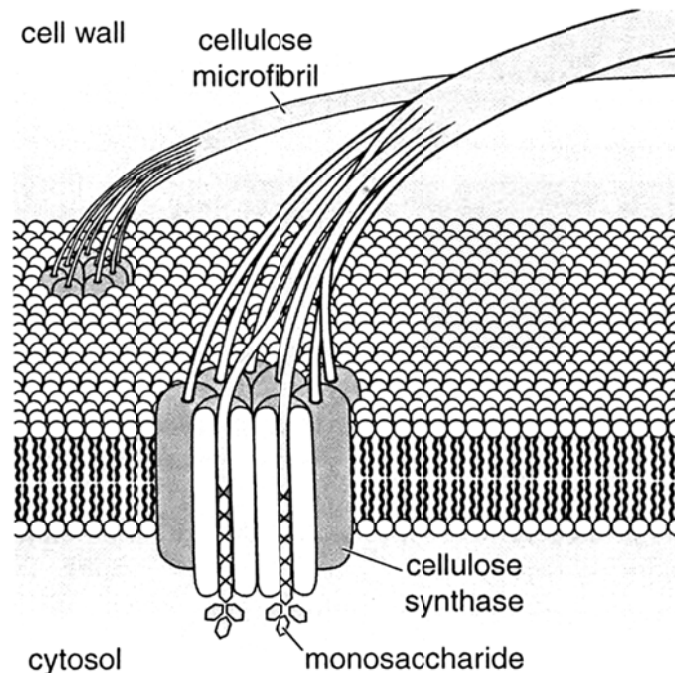


Fig. 1

- (a) (i) Name the monosaccharide units that linked together to form cellulose.
 β -glucose;; [1]
- (ii) Name the bond between monosaccharide in cellulose catalysed by cellulose synthase.
(β - 1,4 -) glycosidic bond;; [1]
- (iii) Describe how the bond in (ii) is formed. You may use an annotated diagram in the space below.
1. *β -glucose w 1 inverted, OH grps at C1 & C4 reacting;;*
 2. *condensation, formation of β -1,4-glycosidic bond;;*
- [2]
- (b) Explain how cellulose synthase speeds up the formation of microfibrils.
1. *(contact residues at) active site (forms temporary bonds) binding β -glucose*
.....
with complementary shape;;
.....
 2. *forms ES cplx*
.....

and lowers E_a ;;

3. cite possible mechanism

e.g. reducing bond angle in substrate OR

e.g. providing favourable microenv ➡ Δ pH / e- distribution;; [3]

(c) Explain why cellulose cannot be synthesised inside the cell.

1. large insoluble molecule

cannot be secreted after fibril assembly;; [1]

Cellulose fibres are made up of many cellulose microfibrils. Each microfibril is made up of many cellulose molecules lying parallel to each other.

(d) With reference to Fig. 1 and your knowledge of the structure of cellulose, explain how cellulose is suited to perform its function.

1. alternative glucose monomer is inverted allowing formation of straight chain

thus bundle to form fibrils to give $\uparrow$ tensile strength;;

2. w protruding OH grps

that can form cross-links;;

3. high mol wt

insoluble;;

4. AVP;; [3]

[Total:11]

- 2 Fig. 2.1 shows ribosomes attached to the endoplasmic reticulum, carrying out insulin synthesis.

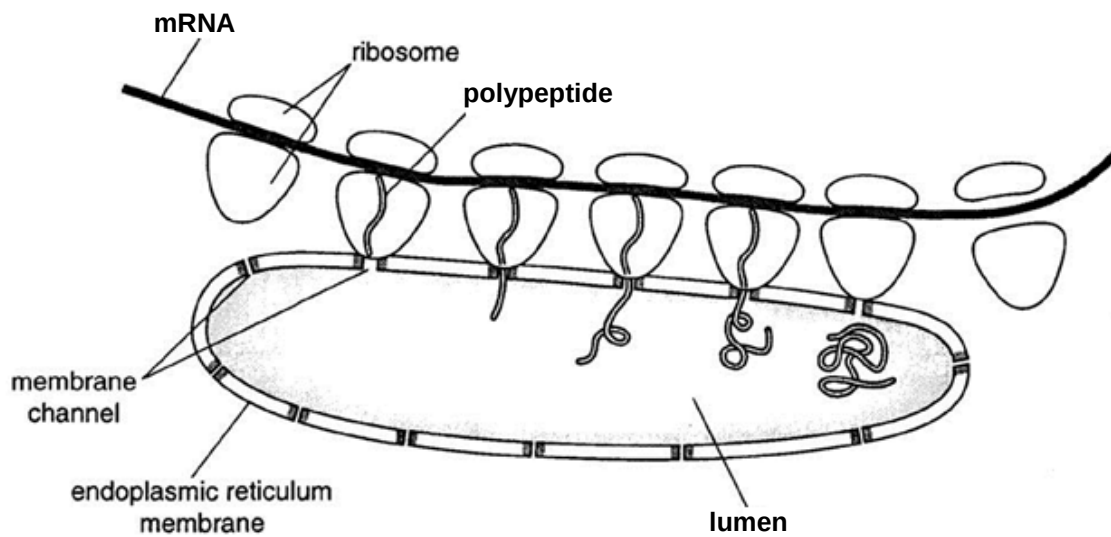


Fig. 2.1

- (a) With reference to Fig. 2.1, describe the function of the membrane channel.

1. *bind signal peptide*

to bring translation initiation cplx to ER;;

2. *provide (hydrophilic) channel for polypep*

to enter lumen;;

[2]

- (b) The polypeptide entering the lumen is processed before it leaves the endoplasmic reticulum. Describe the post-translational processing of the polypeptide to form insulin.

1. *signal peptide cleaved*

from pre-proinsulin → proinsulin;;

2. *C peptide cleaved*

dis bridges formed btw chain A and chain B

proinsulin folded → insulin;;

3. *undergoes glycosylation*

where oligosacc chain is added to polypep;;

[2]

- (c) Insulin synthesised is secreted into the bloodstream. Presence of insulin in the bloodstream lowers blood glucose level. Thus production of insulin is highly regulated.

- (i) Briefly describe the events leading up to the secretion of insulin after its processing in the lumen.

1. *trspt vesicle with insulin fuse with cis face of GA*

pinch off in secretory vesicle at trans face;;

2. *insulin undergoes sorting, packaging & modification by adding, deleting & substituting sugar monomer oligosacc chain*

to form molecular identity tag;;

3. *translocate to/w CSM with aid of cytoskeleton*

vesicle fuse with CSM to release insulin into bloodstream via exocytosis;;

[3]

- (ii) Outline how human insulin can be produced by genetic engineering techniques.

1. *locate insulin gene for chain A & B from cDNA lib*

isolate insulin gene & cut plasmid using same RE;;

2. *mix to allow complementary sticky ends to anneal*

ligate using DNA ligase to form recombinant DNA;;

3. *transform into bact host cell via heat shock*

select recombinant bact using selective medium interacting with selectable marker or blue/white colony screening (using insertional inactivation of lacZ gene);;

4. *grow recomb bact in bioreactor / fermenter with opt cond e.g. nutrients, temp, pH etc.*

lyse bact cells, purify insulin chains;;

5. *cleave β -gal from fusion prot using CNBr*

allow diS bond formation btw chain A & B to form fnal insulin;;

[4]

Fig. 2.2 shows the sequence of bases in a section of the mRNA in Fig. 2.1, coding for the first few amino acids of insulin.

3' AUGUUGCCCCCUUAGGUAU 5'

Fig. 2.2

Each codon codes for a specific amino acid as illustrated in Table 2,

Table 2

UUU Phe	UCU Ser	UAU Tyr	UGU Cys
UUC Phe	UCC Ser	UAC Tyr	UGC Cys
UUA Leu	UCA Ser	UAA End	UGA End
UUG Leu	UCG Ser	UAG End	UGG Trp
CUU Leu	CCU Pro	CAU His	CGU Arg
CUC Leu	CCC Pro	CAC His	CGC Arg
CUA Leu	CCA Pro	CAA Gln	CGA Arg
CUG Leu	CCG Pro	CAG Gln	CGG Arg
AUU Ile	ACU Thr	AAU Asn	AGU Ser
AUC Ile	ACC Thr	AAC Asn	AGC Ser
AUA Ile	ACA Thr	AAA Lys	AGA Arg
AUG Met	ACG Thr	AAG Lys	AGG Arg
GUU Val	GCU Ala	GAU Asp	GGU Gly
GUC Val	GCC Ala	GAC Asp	GGC Gly
GUA Val	GCA Ala	GAA Glu	GGA Gly
GUG Val	GCG Ala	GAG Glu	GGG Gly

- (d) (i) Using information from Table 2, list the corresponding amino acids coded by the mRNA in Fig. 2.2.

met – asp – ser – pro – val - val

[2]

- (ii) Explain why each codon consists of three bases.

1. 20 common aas present thus need minimum of 20 codes;

2. if 1 base ➡ only 4 codes;

if 2 bases ➡ only 16 codes;

thus need 3 bases ➡ 64 codes;

3. allow for stop codes to terminate translation;

[2]

A separate 10 amino acids long section of the polypeptide chain was partially hydrolysed to release oligopeptides shown in Fig. 2.2.

gly-ala-gly-val

gly-gly-lys

gly-val-ile-phe

phe-thr-gly-gly

Fig. 2.2

- (e) (i) With reference to Fig. 2.2, work out the order of amino acids in the polypeptide.

gly-ala-gly-val-ile-phe-thr-gly-gly-lys ;;

[1]

- (ii) In the space provided, illustrate the hydrolysis of a peptide bond between two glycine residues.

1. *illustrate gly-gly dipeptide*
with peptide bond labelled correctly;
2. *water as reactant;*
hydrolysis to form 2 gly;

[2]

- (f) (i) Explain why single base deletions in a coding DNA sequence are more likely to produce non-functional proteins than single base substitutions.

1. *single base deletions → frameshift mutation;*

all aas downstream of mutation will be Δ;

2. *single base sub only affects 1 codon thus 1 aa i.e. missense;*

less Δ to primary struct of prot;

[2]

- (ii) Explain why some base substitutions do not lead to change in the protein coded for.

1. *single base sub at 3rd base of codon*

leading to degenerate code coding for the same aa;;

2. *silent mutation*

due to wobble base phenomenon;;

[2]

[Total: 22]

- 3 The inheritance of a genetic skin disorder in a family is shown in Fig. 3.

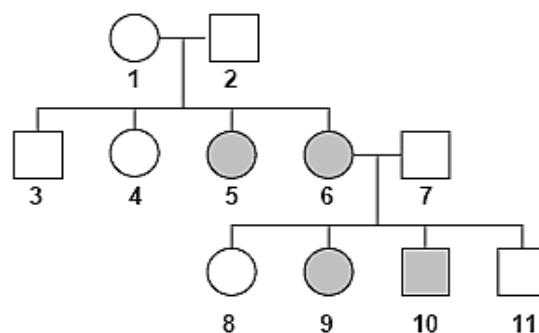


Fig. 3.

(a) With reference to Fig. 3, explain the mode of inheritance of this disease.

1. autosomal recessive;;

2. recessive ➔ skips generation, 5 & 6 has disease but parents ⊗ have

autosomal & not sex-linked ➔ non-affected father (2) has affected daughters (5 & 6);;

[2]

(b) Blood groups of several individuals are listed below.

- Individual 1 – blood group O
- Individual 5 – blood group B
- Individual 6 - blood group A
- Individual 7 - blood group AB

(i) State the blood group individual 2.

AB ® $I^A I^B$

[1]

(ii) State the probability that individual 4 has blood group A.

0.5

[1]

(c) Using appropriate symbols, illustrate the probability that individual 6 and 7 will have a child with blood group A who suffers from the skin disorder.

Legend

D – dominant allele for normal skin

d – recessive allele for skin disorder

I^A – dominant allele for blood grp A

I^B – dominant allele for blood grp B

I^O – recessive allele for blood grp O

	Individual 6	Individual 7
parent phenotype	Bld grp A w skin disorder	Bld grp AB w normal skin
parent genotype	$dd I^A I^O$	$Dd I^A I^B$
	≠	≠
gametes	$d I^A$ $d I^O$	$D I^A$ $D I^B$ $d I^A$ $d I^B$
Punnett Sq		
offspring genotypic ratio		
offspring phenotypic ratio		

probability = 0.25

[4]

[Total: 7]

- 4 (a) Explain, with appropriate examples, how natural selection bring about evolution. [7]
1. *Ref to genetic variation in population;; (e.g. ancestral popn e.g. finches from South America mainland with genetic variation)*
 2. *Ref to selection pressure;; (e.g. occupy diff habitats on Galapagos islands with diff envtal conditions e.g. food source which act as selection pressure)*
 3. *Ref to selective advantage for individuals with adaptations;; (e.g. finches with favourable phenotype e.g. beak shape that can access nectar are naturally selected for)*
 4. *Ref to high survival and reproduction rate, passing down alleles for adaptations to off spring;; (e.g. experience higher survival rate & repro rate pass down alleles coding for beak shape to offspring)*
 5. *cause freq of allele to increase in the gene pool;;*
 6. *Ref to leading to offspring having diff phenotype (e.g. beak shape) and descend with modification;;*
 7. *rapid speciation due to geographical isolation adaptive radiation;;*
- (b) Explain how homology supports Darwin's theory of natural selection. [7]
1. *closely related species sharing recent common ancestor;;*
 2. *have anatomy / morphology / embryological devt / DNA seq of similar basic structure / seq;;*
 3. *relevant example, e.g. same 5 sets of bones in pentadactyl limb;;*
 4. *but spp show descent with modification*
 5. *due to natural selection by diff selection pressures in diff env;;*
 6. *e.g. pentadactyl limb evolve to adapt to diff trspt medium;;*
 7. *e.g. fins in whales to swim in marine habitat;;*
 8. *e.g. wings in bats to fly in aerial habitat;;*
 9. *AVP;;*
- (c) Describe the sources of genetic variation in a population. [6]
1. *mutation is the only source of new alleles;;*
 2. *by gene / point mutation, chromosomal aberrations;;*
 3. *sexual repro leads to diff allelic combinations from male & female parent*
 4. *due to random fertilisation;;*
 5. *requiring genetically varied haploid gametes formed by meiosis;;*
 6. *giving rise to combination of diff alleles due to crossing over / random segregation / independent assortment;;*
 7. *AVP;;*
- [Total: 20]
- 5 (a) Describe how the Calvin cycle differs from the Krebs cycle. [6]
1. *ref metabolic process - photosynthesis, (aerobic) respiration;;*
 2. *ref nature of rxn - anabolic, catabolic*

3. *ref location - stroma of chloroplast, matrix of mitochondria;;*
4. *ref coenz - NADP, NAD & FAD;;*
5. *ref role of CO₂ – used in C fixation, released in oxidative decarboxylation;;*
6. *ref role of ATP – used in GP reduction & RuBP regeneration, produced via substrate level phosphorylation;;*
7. *ref initial S – RuBP + CO₂ , acetylCoA + OAA;;*
8. *AVP;;*

(b) Explain the small yield of ATP under anaerobic conditions in both yeast and mammals. [8]

1. *absence of O₂ under anaerobic conditions*
 ➔ *absence of final e- acceptor in ETC;;*
2. *e- carriers remained reduced*
 NAD⁺ & FAD⁺ not regenerated;;
3. *Krebs cycle cease to fn*
 pyruvate unable to enter link rxn, Krebs;;
4. *ATP not produced via oxidative phosphorylation / chemiosmosis;*
5. *pyruvate → ethanol + CO₂ in yeast*
 pyruvate → lactate in mammals;;
6. *NADH from glycolysis oxidised / regenerated to NAD⁺*
 NAD⁺ enters / recycled to sustain glycolysis;;
7. *producing 2 net ATP per glucose*
 via substrate level phosphorylation;;
8. *only small yield of ATP cf to 38 ATP in oxidative phosphorylation*
 as much of the chem en remains trapped w/in ethanol & lactate;;
9. *evaporate in ethanol in yeast*
 converted back to pyruvate in liver of mammals;;
10. *AVP;;*

(c) Explain how the structure of mitochondria and chloroplasts supports the theory that these organelles are of prokaryotic origin. [6]

1. *ref own DNA / genome / chromosome*
 circular in nature similar to bact;;
2. *ref DNA packed in non-histone prot similar to bact;;*
3. *ref diff in genetic code from nuclear genes;;*
4. *ref double mem*
 phospholipids in outer mem similar to eukaryote, inner mem similar to bact;;
5. *ref presence of 70S ribosome similar to bact*
 instead of 80S found in cytosol;;

6. *endosymbiotic theory*

organelles are prokaryotes engulfed by eukaryotic host;;

[Total: 20]